packages/graph-view/src/utils/__tests__/layerStats.test.ts

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Overview

A new test file `packages/graph-view/src/utils/__tests__/layerStats.test.ts` was added. It imports `vitest`, `computeLayerStats`, and the `GraphNode`/`Layer` types (R1-R3). Helper constructors `node`, `layer`, and `indexById` are defined (R5-R30). The file contains a `describe` block with several `it` tests (R32-R118).

Key changes

- **Imports**: `vitest`, `computeLayerStats`, and type imports added (R1-R3).
- **Helpers**:
  - `node` creates a `GraphNode` (R5-R17).
  - `layer` creates a `Layer` (R19-R26).
  - `indexById` builds a `Map<string, GraphNode>` (R28-R30).
- **Test cases**:
  - Resolved node counting only existing nodes (R32-R38).
  - Complexity aggregation thresholds at 30 % for `simple`, `complex`, and `moderate` (R40-R86).
  - Preference of `complex` over `moderate` when both exceed the threshold (R64-R75).
  - Empty layer treated as `simple` with `resolvedCount` = 0 (R88-R92).
  - Performance guard: 100 layers × 100 nodes, execution under 50 ms (R94-R118).

Impact

- Provides unit coverage for `computeLayerStats`, ensuring correct filtering and aggregation logic.
- Adds a regression guard for algorithmic complexity, comparing the new $O(N + \Sigma K_i)$ path against the previous $O(N \times K \times L)$ approach.
- Centralizes test helpers, reducing duplication across tests.

Risks & follow-ups

- The performance test depends on `performance.now()`; CI hardware variability could cause flakiness.
- If `computeLayerStats` is refactored, these tests may fail; review threshold logic to keep it consistent.
- The 30 % threshold is hard-coded in the tests; confirm that this value is documented and not changed elsewhere.